

CHAPTER 3: METHODOLOGY (CORRECTED & ALIGNED VERSION)

Colegio de Montalban — Integrated Library Management System (CDM ILMS)

3.1 Research Design

This study adapts a **descriptive-evaluative research design** in the development and evaluation of the proposed *Integrated Multi-Platform Library Management System with Mobile Book Preview and Reservation* for Colegio de Montalban (CDM). Descriptive research is selected because it enables systematic collection, analysis, and quantitative evaluation of system acceptability across software quality characteristics.

The system evaluation adheres to the **ISO/IEC 25010 Software Product Quality Model**, assessing six (6) core quality attributes: Functional Suitability, Performance Efficiency, Usability, Reliability, Security, and Compatibility. The software development process follows the **Agile Software Development Life Cycle (SDLC)** methodology, which supports iterative prototyping, continuous integration, and feedback-driven refinement.

3.2 System / Project Overview

The project is an *Integrated Multi-Platform Library Management System (CDM ILMS)* featuring a Single-Page Application (SPA) Desktop Librarian Management System and a Progressive Web Application (PWA) Mobile Student Portal. The system centralizes academic reference cataloging across Colegio de Montalban institutes, including the Institute of Computer Studies (ICS), Institute of Teacher Education (ITE), and Institute of Business and Entrepreneurship (IBE).

Core features include: (1) Standardized MARC 21 academic cataloging with Library of Congress Call Numbers; (2) Real-time multi-criteria searching and memoized filtering (`useMemo`, `K` shortcut); (3) Automated borrowing and return transaction ledgering with real-time stock inventory control (`available`/`total`); (4) Student self-service mobile PWA with offline caching and mobile reservations; (5) Automated 6-digit Email OTP account password recovery; and (6) Dynamic 2D QR Code receipt canvas generation for turnstile verification.

3.3 Respondents of the Study

The respondents of the study consist of selected active student borrowers and library administrative staff at Colegio de Montalban. Student respondents evaluate the usability, accessibility, and responsiveness of the Mobile PWA Portal, while library staff evaluate inventory cataloging, circulation processing, overdue tracking, and reporting efficiency.

3.4 Sampling Technique

This study utilizes **Purposive Sampling**. Purposive sampling is selected because evaluating specialized library management software requires respondents who actively interact with library operations and academic resource materials. Selecting frequent student library patrons and active library staff ensures that gathered evaluation data is highly reliable and representative of true operational workflows.

3.5 Instruments and Methods for Data Collection

Data collection relies on survey questionnaires, direct user observation, and system transaction logs. The primary evaluation instrument is an **ISO/IEC 25010 Software Quality Survey Questionnaire** rated on a 5-point Likert scale (5 - Strongly Agree to 1 - Strongly Disagree).

3.6 Hardware Requirements

- **Development Workstation:** Laptop / Desktop Computer (Intel Core i5/i7 or Apple Silicon, 8GB+ RAM) for Node.js API and React SPA development.
- **Mobile Testing Devices:** Android & iOS Mobile Smartphones for testing Mobile PWA responsiveness, touch controls, and offline Service Worker caching.
- **Barcode & Turnstile Scanners:** Optical Camera / 2D Scanner devices for verifying QR Code student receipts at library checkpoints.

3.7 Software Specifications & Technology Stack (Corrected)

The software specifications reflect the actual production technology stack utilized in building CDM ILMS:

Component Layer	Software Specification / Technology	Functional Role & Purpose
Desktop Frontend SPA	React 19, TypeScript, Vite 6, Tailwind CSS v4, Lucide React	Single-Page Application (SPA) for librarians; MARC 21 filtering, 60 FPS `useMemo` search.
Mobile PWA Portal	Progressive Web App (PWA), Service Worker, Manifest JSON	Mobile student self-service portal, offline caching, mobile borrow reservations.
Backend REST API	Node.js (v25+), Express.js (v5), CORS Middleware	Asynchronous RESTful API server handling business logic, routes, and JSON responses.
Database Layer	SQLite 3, `better-sqlite3` driver	Synchronous embedded relational database (`library.db`) managing tables & transaction logs.
Security Engine	JSON Web Token (JWT), `bcrypt` (10 rounds)	Stateless session authentication and salted password hash security.
QR Verification	`qrcode` Canvas Engine	Dynamic 2D QR Code receipt canvas generation for turnstile verification.